

SVKM'S NMIMS
MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT & ENGINEERING

Academic Year: 2022-2023

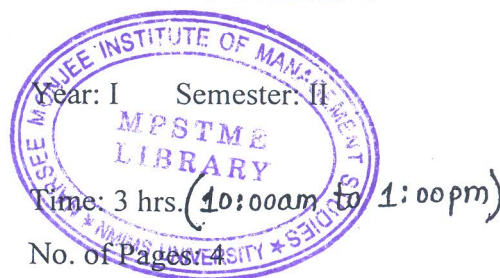
Program: MCA

Stream: Computer

Subject: Software Engineering

Date: 02/05/2023

Marks: 100



Final Examination

Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

- 1) Question No. 1 is compulsory.
- 2) Out of remaining questions, attempt any 4 questions.
- 3) In all 5 questions to be attempted.
- 4) All questions carry equal marks.
- 5) Answer to each new question to be started on a fresh page.
- 6) Figures in brackets on the right hand side indicate full marks.
- 7) Assume Suitable data if necessary.

Q1		Answer briefly:	[20]
CO-1; BL-III	a.	Identify if the given statement is a software myth or not. Justify your answer. Also mention the category if falls into. i) "If I decide to outsource the software project to a third party, I can just relax and let that firm build it." ii) "A general statement of objectives is sufficient to begin writing programs-we can fill in the details later"	[5]
CO-4; BL-IV	b.	Compare White Box and Black Box testing.	[5]
CO-2; BL-II	c.	Explain the following two architectural styles i) Data Centered Architectures ii) Data Flow Architectures	[5]
CO-3; BL-II	d.	Explain UML state chart diagrams. Provide description of the components of a state chart diagram and how they are used to model the behavior of a system.	[5]

Q2 CO-1; BL-IV	a.	Analyze the given scenario identify the appropriate software process model for it & draw the diagram of that model. A software development company is tasked with developing a new e-commerce platform for a client. The client has a basic understanding of what they want the platform to do, but they are not sure about all of the details. The client wants to be able to see progress and provide feedback as the development progresses. Choose Appropriate Prescriptive Model.	[10]
Q2 CO-4; BL-V	b.	<pre> Begin int x,y, power; float z; input(x,y); if(y<0) power = -y; else power = y; z=1; while (power!=0) {z=z*x; power = power - 1;} if(y<0) z=1/z; output(z); end For the above code - a) Define Cyclometric Complexity b) Draw flow graph for the above program c) Determine Cyclometric Complexity </pre>	[10]
Q3 CO-3; BL-VI	a.	Design an Activity diagram for Online Food Delivery App- KitchenCorner, which delivers food to the customer's doorstep directly from the restaurants. Admin coordinates and manages with different food outlets. Customers can register and request food Item to the doorstep. Customer can add food items from only 1 Outlet to the cart for a single order and make the payment through netbanking, Gpay or COD.	[10]

Q3 CO-2; BL-VI	b.	Discuss the various requirement engineering tasks in detail.	[10]
Q4 CO-2; BL-VI	a.	DFD for a Social Networking site The Absolute Beginners Inc. is planning to launch a revolutionary social networking site, EyeCopy. You have been entrusted with designing a DFD for the proposed application. In particular, you have been asked to show the following scenarios: • User registration • User login • Profile update Design different levels of DFD to depict the above data flow and the corresponding processes. Should there be any data store in the DFD?	[10]
Q4 CO-1; BL-II	b.	Mention Agile process. Explain SCRUM model with diagram also discuss its key features.	[10]
Q5 CO-4; BL-II	a.	Discuss Unit Testing. State the use of Stubs and Drivers.	[10]
Q5 CO-2; BL-II	b.	Explain different steps involved for mapping data flow into software architecture with suitable example.	[10]
Q6 CO-3; BL- VI	a.	Consider the given scenario You are designing a system for an online learning platform that allows students to enroll in courses and interact with instructors and other students. Design a use case diagram that illustrates the main functionalities of the system. Your diagram should include the actors and use cases involved in the system, as well as their relationships. Explain briefly the purpose of each use case.	[10]
Q6 CO-1; BL- II	b.	Explain different CMMI levels.	[10]

Q7 CO-4; BL-II		Write short notes on: 1) McCall's Software Quality Factors 2) SQA Activities 3) Golden Rules for User Interface Design 4) Architectural Design Level	[20]
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